

NOTE:

- These are not exact midterm questions. These questions are intended to review the material.

1 Comparing Functions

For each of the following pairs of functions, write down the symbol that most precisely describes the asymptotic relation between $f(n)$ and $g(n)$; i.e., if $f(n) = x(g(n))$ where $x \in \{o, O, \Theta, \Omega, \omega\}$, then write down x under the "Relation" column in the corresponding row (an example is provided on the 0th row). You do not need to justify your answer (but try one for practice). Make sure to clearly distinguish O and o .

#	$f(n)$	$g(n)$	Relation
0	$n^4 + 3$	$3n^4$	Θ
1	$n^{0.5}$	n^{-4}	
2	$\log(n^4)$	$\log(n^{0.5})$	
3	$4n^3$	$3n^4$	
4	$n^{-3} + 0.5$	4	
5	$4^n + 3$	$3^n + 4$	

2 Solving Recurrences

Provide a tight asymptotic solution for each of the following recurrences. If Master Method applies, also state the case of the master method that applies. If it does not, put "N/A". The first row is provided as an example. You do not need to justify your answer (but try one for practice).

Recurrence	Solution	Case
$T_0(n) = 9T(n/9) + \Theta(n)$	$\Theta(n \log n)$	2
$T_1(n) = 3T_1(n/9) + \Theta(n)$		
$T_2(n) = 9T_2(n/3) + \Theta(n)$		
$T_3(n) = 9T_3(n/3) + \Theta(n^2)$		
$T_4(n) = T_4(n - 1000) + \Theta(n^2)$		
$T_5(n) = T_5(n/3) + \Theta(1)$		

3 Create another Algorithm

Tom has a set, A , of n nuts and a set, B , of n bolts, such that each nut has a unique matching bolt. Unfortunately, the nuts in A all look the same, and the bolts in B all look the same, as well. The only comparison that Tom can make is to take a nut-bolt pair (a, b) , such that a is in A and b is in B , and test if the threads of a are larger, smaller, or a perfect match with the threads of b . Describe an efficient algorithm for Tom to match up all of his nuts and bolts.